HLX01-POS/SPOS



ARMOR-X™ and ARMOR-X*tra*™ HL

Instrumentation Cable

600 Volt Type MC-HL Cable

PVC/Nylon Insulated Singles, Sizes 18 AWG or 16 AWG, 90°C

Continuous Corrugated Aluminum Armor with -40°C PVC Jacket

APPLICATIONS

Suitable for use as follows:

- Southwire's 600 Volt ARMOR-X™ and ARMOR-X*tra*™ Type MC-HL POS/SPOS Instrumentation Cable provides an impervious sheath recommended as an economical alternative to traditional conduit systems
- For use on Class 1 remote-control and signaling circuits where 600 volts is desired
- For use indoors, outdoors, direct burial, encased in concrete, open trays, troughs or continuous rigid cable supports
- For use in Classes I, II and III Division 1 and 2 hazardous locations
- For use as a nonpower-limited fire alarm circuit cable (NPLF) per NEC Article 760. Rated for wet and dry applications at temperatures not to exceed 90°C

SPECIFICATIONS

Southwire's 600 Volt ARMOR-X™ and ARMOR-X*tra*™ Type MC-HL Instrumentation Cable is manufactured and tested in accordance with the latest revisions of:

- UL 83 – Thermoplastic Insulated Wire
- UL 1277 – Electrical Power and Control Cables
- UL 1569 – Metal-Clad Cables
- UL 1309 – Listed as Marine Shipboard Cable
- UL 2225 – Safety Metal-Clad Cables and Cable-Sealing for Use in Hazardous (Classified) Locations
- IEEE 1202 – Flame Testing of Cables for Use in Cable Tray in Industrial and Commercial Occupancies (70,000 BTU/hr)
- ICEA T-29-520 – Vertical Cable Tray Flame Tests (210,000 BTU/hr)

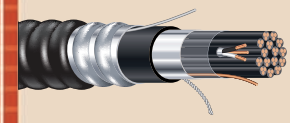
CONSTRUCTION

- Southwire's 600 Volt ARMOR-X™ and ARMOR-X*tra*™ Type MC-HL continuous corrugated aluminum armor provides an aluminum sheath impervious to moisture, liquids and gases
- Cable is flame retardant UL-listed for cable tray use, direct burial, sunlight resistant and minus 40°C installation temperature
- Conductors are sizes 18 AWG or 16 AWG, 7-strand copper conductors, PVC insulation with nylon covering, color coded, twisted pairs and triads, group of pairs and triads, numeric print identification on the groups, aluminum polyester foil with 100% coverage, overall tinned drain wire, a nylon ripcord and a black PVC jacket
- The paired conductors are colored black, red and numbered while the traids are color coded black, red and numbered
- **Scope:** This specification covers multiconductor PVC insulated, continuous corrugated aluminum armored, thermoplastic jacketed, Type MC-HL 600 volt instrumentation cable for use indoors, outdoors, direct buried, encased in concrete, open trays, troughs or continuous rigid cable supports. For use on Class 1 remote control and signaling circuits where 600 volts is desired. For use in Classes I, II, and III Division 1 and 2 hazardous locations. For use as a nonpower-limited fire alarm circuit cable (NPLF) per NEC Article 760. This cable is capable of operating continuously at a conductor temperature of 90°C in wet or dry locations

Southwire® OEM Division

UL LISTED

MADE IN USA



WEIGHTS, MEASUREMENTS AND PACKAGING

SIZE	NUMBER OF		INSULATION THICKNESS		INSULATION JACKET THICKNESS		INNER JACKET THICKNESS		APPROX. CORE DIAMETER		APPROX. ARMOR-X DIAMETER		OUTER JACKET THICKNESS		APPROX. OVERALL DIAMETER		APPROX. NET WEIGHT	
AWG OR kcmil	PAIRS	TRIADS	inches	mm	inches	mm	inches	mm	inches	mm	inches	mm	inches	mm	inches	mm	lbs/ Mft	kg/ km
HLX01-POS SINGLE PAIR AND TRAID CABLE (POS)																		
16 (7)	1	-	0.015	0.38	0.005	0.13	0.045	1.14	0.302	7.67	0.520	13.2	0.060	1.5	0.64	16.3	165	246
16 (7)	-	1	0.015	0.38	0.005	0.13	0.045	1.14	0.317	8.05	0.520	13.2	0.050	1.3	0.62	15.7	171	254
HLX01-SPOS MULTI-PAIR AND TRAID CABLES (SPOS)																		
18 (7)	2	-	0.015	0.38	0.005	0.13	0.045	1.14	0.437	11.1	0.640	16.3	0.060	1.5	0.76	19.3	224	333
16 (7)	2	-	0.015	0.38	0.005	0.13	0.045	1.14	0.416	10.6	0.640	16.3	0.060	1.5	0.76	19.3	249	371
18 (7)	4	-	0.015	0.38	0.005	0.13	0.045	1.14	0.516	13.1	0.740	18.8	0.060	1.5	0.86	21.8	294	438
16 (7)	4	-	0.015	0.38	0.005	0.13	0.045	1.14	0.552	14.0	0.780	19.8	0.060	1.5	0.90	22.9	365	543
18 (7)	8	-	0.015	0.38	0.005	0.13	0.045	1.14	0.676	17.2	0.910	23.1	0.060	1.5	1.03	26.2	443	659
16 (7)	8	-	0.015	0.38	0.005	0.13	0.045	1.14	0.720	18.3	0.910	23.1	0.060	1.5	1.03	26.2	528	786
18 (7)	12	-	0.015	0.38	0.005	0.13	0.045	1.14	0.814	20.7	1.050	26.7	0.060	1.5	1.17	29.7	569	847
16 (7)	12	-	0.015	0.38	0.005	0.13	0.045	1.14	0.868	22.0	1.220	31.0	0.060	1.5	1.34	34.0	744	1107
18 (7)	16	-	0.015	0.38	0.005	0.13	0.045	1.14	0.944	24.0	1.220	31.0	0.060	1.5	1.34	34.0	746	1110
16 (7)	16	-	0.015	0.38	0.005	0.13	0.045	1.14	1.023	26.0	1.350	34.3	0.060	1.5	1.47	37.3	944	1450
18 (7)	24	-	0.015	0.38	0.005	0.13	0.045	1.14	1.165	29.6	1.470	37.3	0.060	1.5	1.59	40.4	995	1481
16 (7)	24	-	0.015	0.38	0.005	0.13	0.045	1.14	1.218	30.9	1.540	39.1	0.070	1.8	1.68	42.7	1288	1917
18 (7)	36	-	0.015	0.38	0.005	0.13	0.045	1.14	1.276	32.4	1.590	40.4	0.070	1.8	1.73	43.9	1324	1970
16 (7)	36	-	0.015	0.38	0.005	0.13	0.045	1.14	1.404	35.7	1.670	42.4	0.070	1.8	1.81	46.0	1703	2534
16 (7)	-	4	0.015	0.38	0.005	0.13	0.045	1.14	0.636	16.2	0.870	22.1	0.060	1.5	0.99	25.1	453	674
16 (7)	-	8	0.015	0.38	0.005	0.13	0.045	1.14	0.838	21.3	1.050	26.7	0.060	1.5	1.17	29.7	738	1098
16 (7)	-	12	0.015	0.38	0.005	0.13	0.045	1.14	1.023	26.0	1.350	43.3	0.060	1.5	1.47	37.3	1008	1500

POS – single or multiple pairs or traids with an overall shield.
SPOS – multiple shielded pairs or traids with an overall shield.

- **Standards:** The following standards will form part of this specification - ASTM B8, UL 83, UL 1569, and UL 1277
- **Conductors:** The conductor will be 18 AWG or 16 AWG, 7-strand, bare copper conductor
- **Insulation:** The insulation will be PVC with a nylon covering meeting the requirements of the referenced standards. The insulation thickness will be 0.015 inches of PVC with 0.005 inches of nylon. Individual conductors will be color coded with the paired conductors colored black, red and numbered while the triads are color coded black, blue, red and numbered
- **Assembly:** The insulated conductors will be twisted pairs or triads, group of pairs or triads, numeric print identification on the groups, aluminum polyester foil with 100% coverage and tinned copper drain wire over each group, overall aluminum polyester foil with 100% coverage, overall tinned drain wire, a nylon ripcord, and a black minus 40°C PVC jacket
- **Armor:** Continuous, impervious, welded, corrugated copper free aluminum armor will be applied over the assembly
- **Jacket:** The cable will be covered with a black PVC jacket conforming to the requirements specified for polyvinyl chloride in ICEA. The average thickness will be in accordance with ICEA, and the minimum spot thickness will be not less than 80% of the average thickness. The jacket will be sunlight resistant and will meet the requirements of the IEEE 1202 (70,000 BTU/hr) and ICEA T-29-520 (210,000 BTU/hr) vertical cable tray flame tests. It is suitable for use at a minimum ambient temperature of minus 40°C
- **Identification:** A manufacturer's identification will be printed on the jacket
- **Tests:** Physical and electrical tests will be conducted in accordance with the requirements of the referenced standards